

Books to Scrape

Data Science PGC Capstone Project

1. Problem Statement

The project focuses on collecting book data from the Books to Scrape website and analysing it to understand book prices, ratings, and availability.

The collected data is cleaned and explored to find useful patterns. A simple machine learning model is also used to identify whether a book is highly rated based on its price.

2. Objectives

- Collect 1,000 book records using web scraping.
- Clean and prepare the collected data for analysis.
- Analyse book prices, ratings, and availability.
- Build a simple machine learning model to identify highly rated books.
- Create a Power BI dashboard to present the findings.

3. Data Source & Web Scraping

Data Source: Books to Scrape

Website: Books to Scrape

Records Collected: 1,000 books

Fields Collected:

- Title
- Price
- Availability
- Rating

Method Used: Python web scraping using Requests and BeautifulSoup.

The data was collected from 50 pages of the website and saved as Books_Raw_Data.csv for further cleaning and analysis.

4. Data Cleaning & Preprocessing

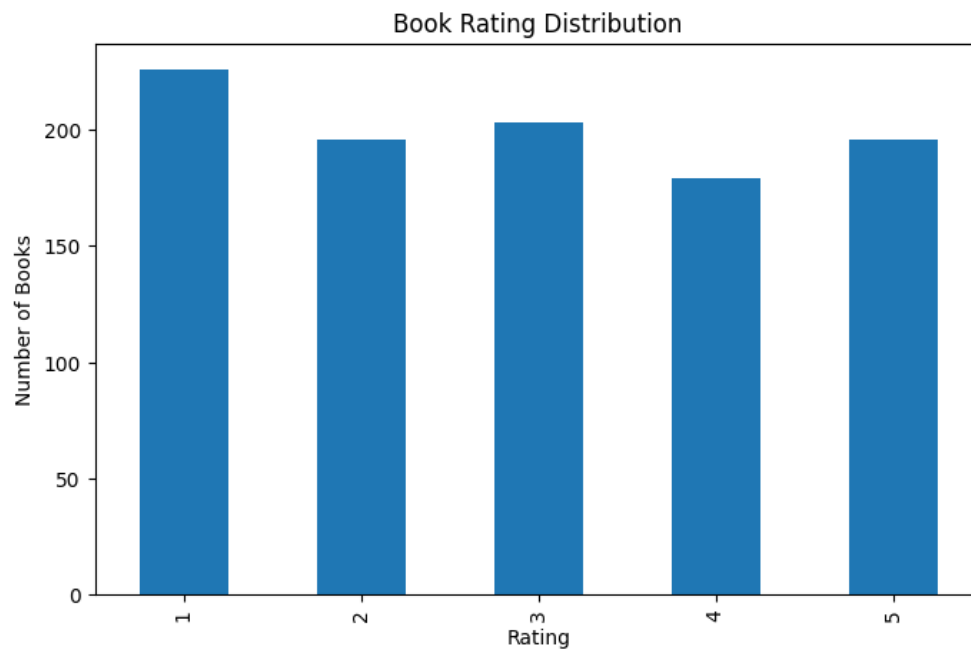
After collecting the data, I checked it and made some changes so that it could be used for analysis.

- I checked for missing and duplicate values.
- Duplicate records were removed.
- The price was converted into a numeric format.
- Book ratings were changed from words to numbers from 1 to 5.
- I created price categories such as Low, Medium, High and Very High.
- I also created rating categories based on the ratings.
- The cleaned dataset was saved as Books_Cleaned_Data.csv.

5. EDA & Data Analysis

After cleaning the data, I used a few charts to understand the books better. The first chart shows how the ratings are distributed among the 1,000 books.

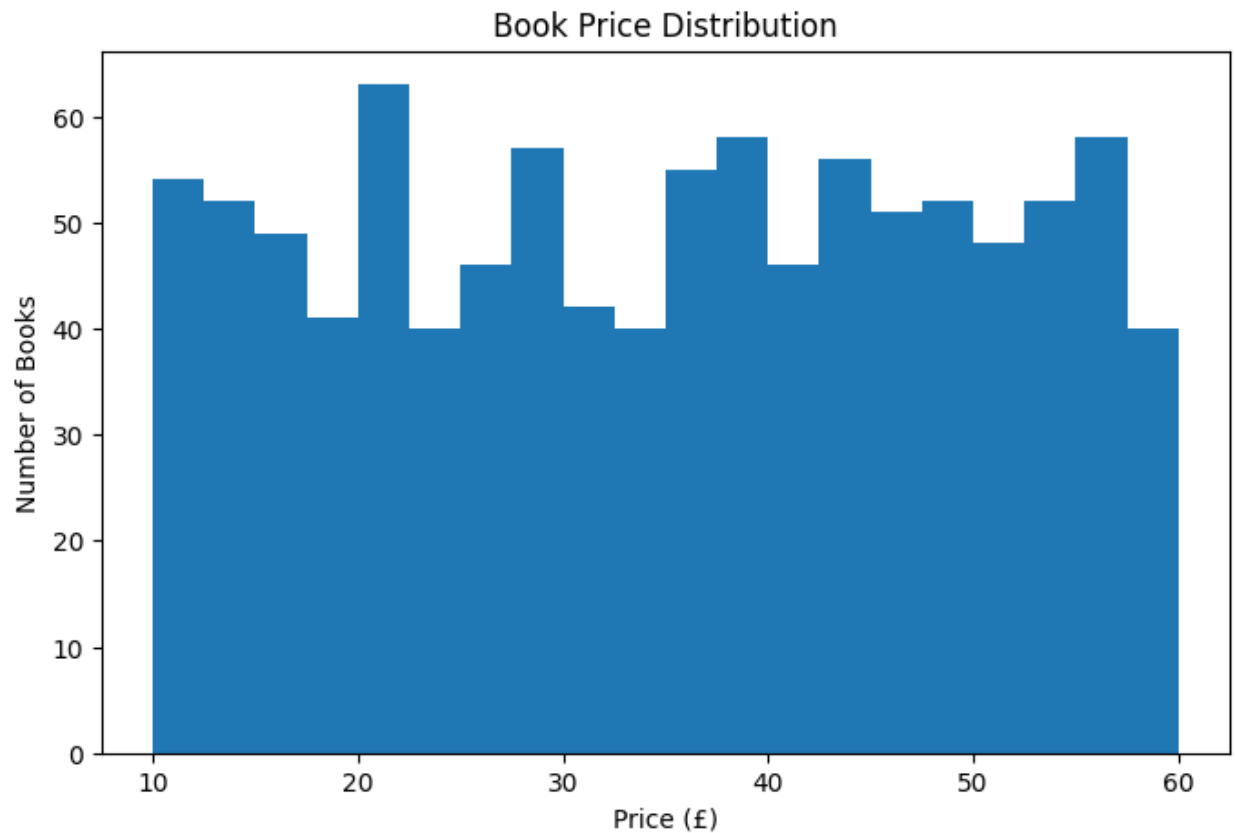
Book Rating Distribution



Insight:

The ratings are spread across 1 to 5 stars, with some ratings appearing more frequently than others. This shows that the books on the website have different levels of customer ratings.

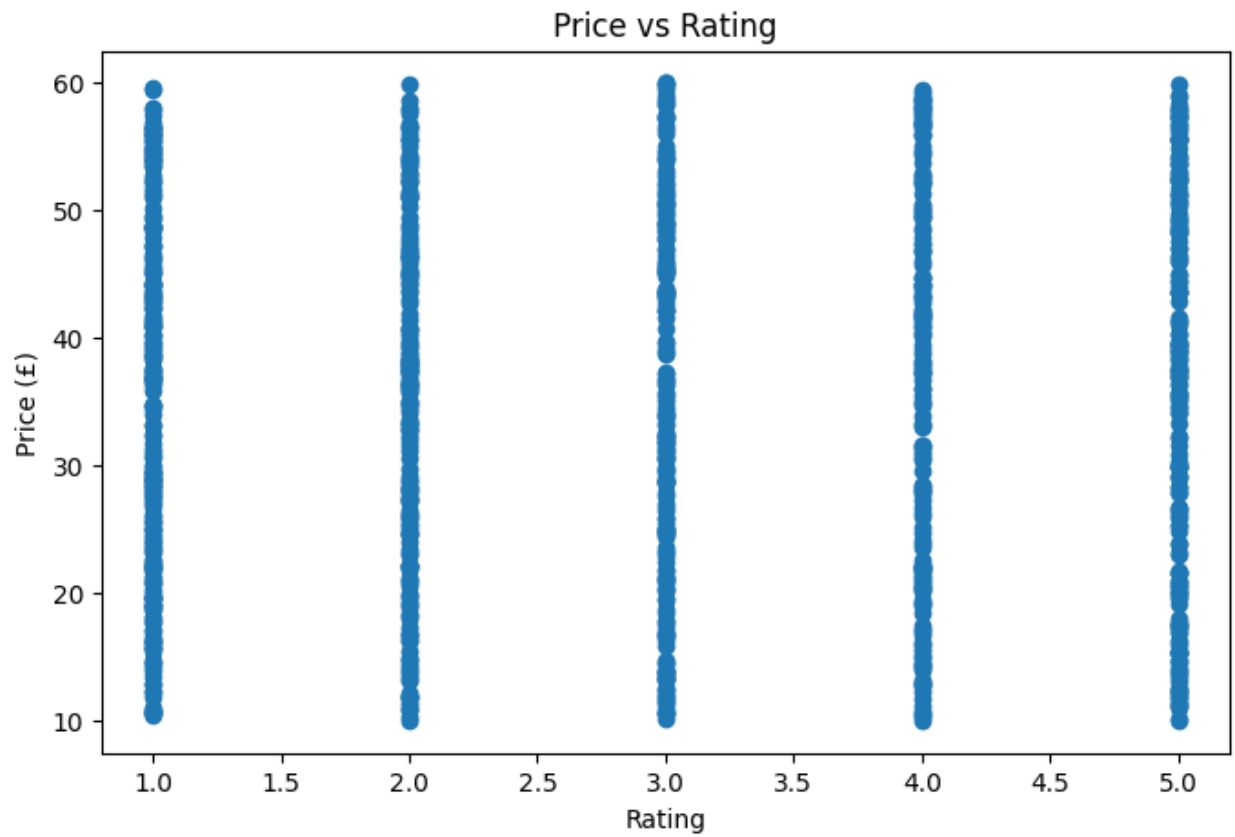
Book Price Distribution.



Insight:

The chart shows how the prices of the 1,000 books are distributed across different price ranges. It helps to understand which price ranges contain more books and how widely the prices vary.

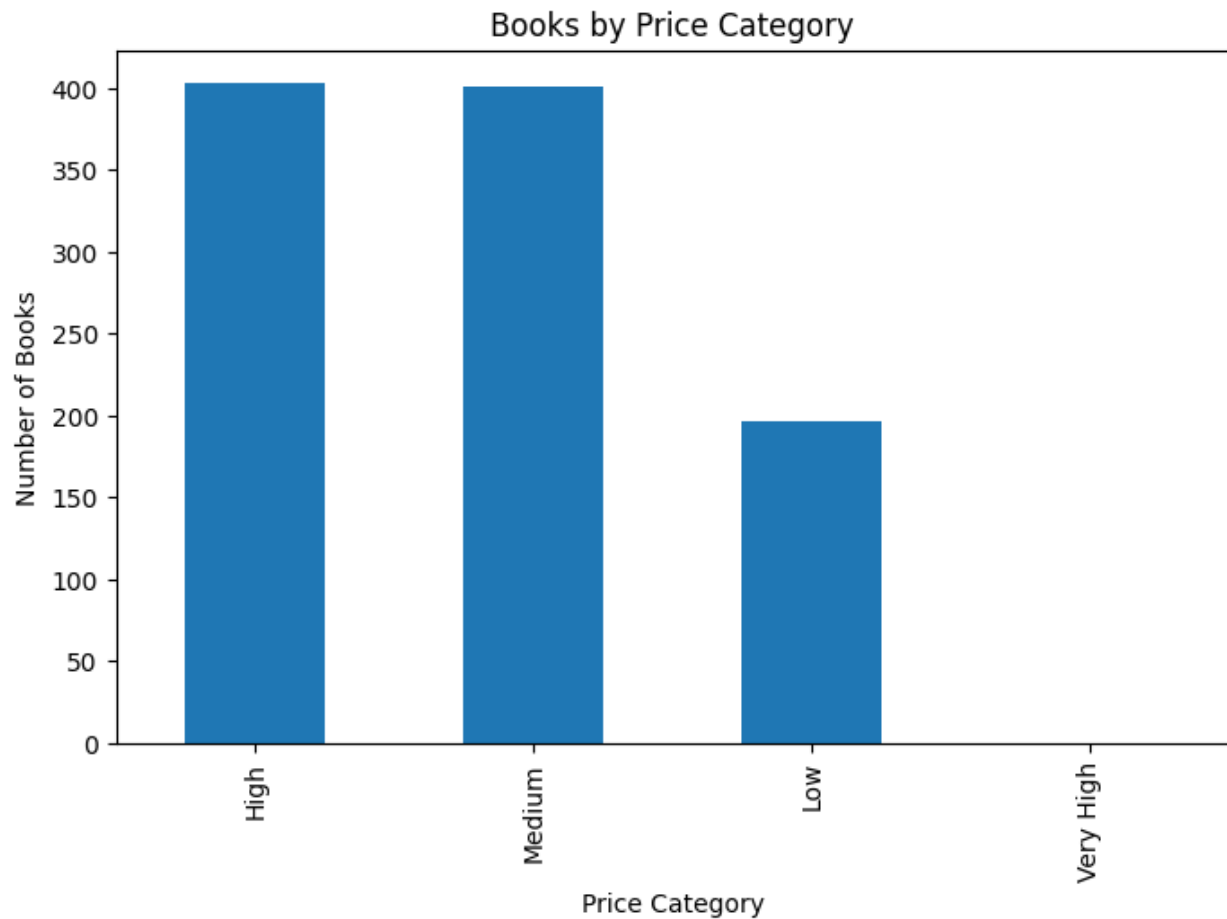
Price vs Rating



Insight:

This chart shows how the prices of the books are spread across different price ranges. It helps in understanding the common price range of the books in the dataset.

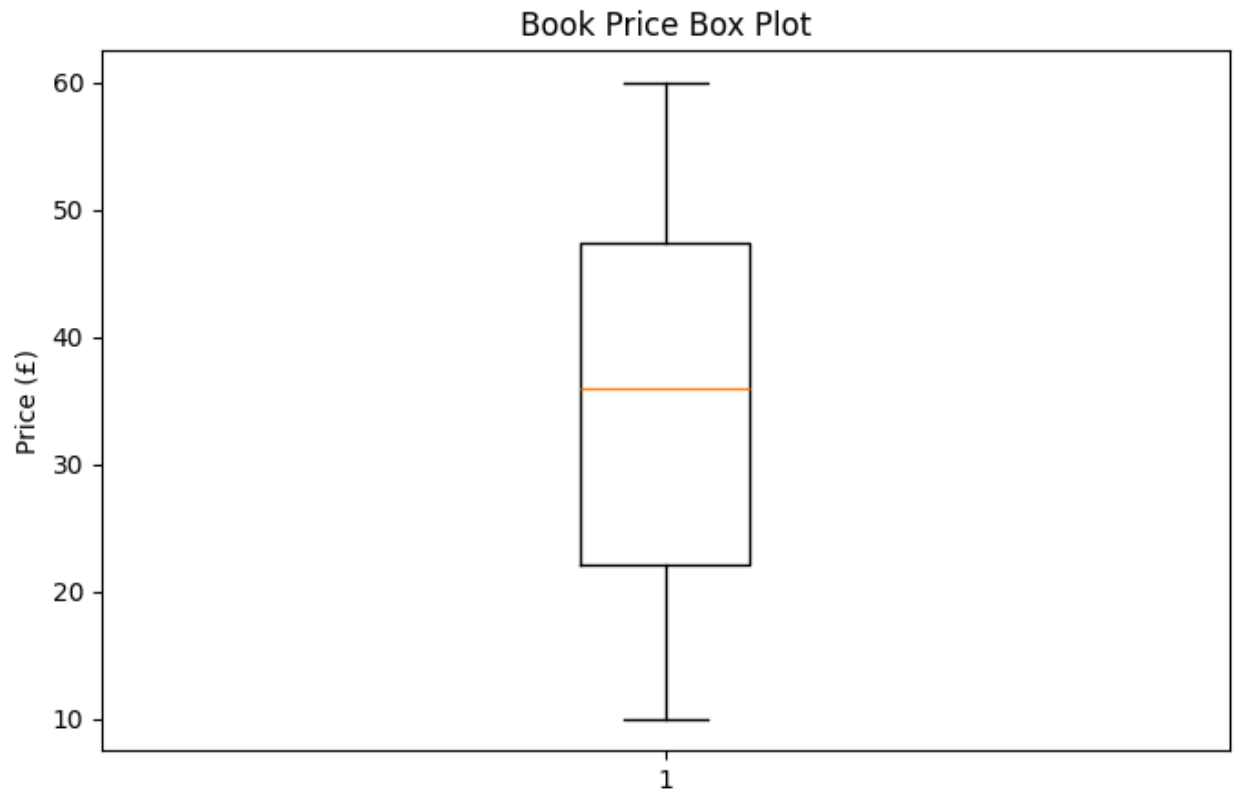
Books by Price Category



Insight:

This chart shows how the 1,000 books are divided into different price categories such as Low, Medium, High and Very High. It helps to understand which price range contains the largest number of books in the dataset.

Book Price Box Plot



Insight:

The box plot shows the spread of book prices and helps identify the middle price range and any unusually high or low prices in the dataset.

6. Key Findings from EDA

From the exploratory data analysis, I found the following points:

- The dataset contains 1,000 books with different ratings and prices.
- Book prices are spread across different price ranges.
- The ratings range from 1 to 5 stars.
- The price category analysis shows the distribution of books across different price groups.
- The Price vs Rating chart helps compare book prices with their ratings.

7. Machine Learning Model

For the machine learning part, I used a Decision Tree Classifier to identify whether a book is highly rated based on its price.

Target Variable:

High_Rating — 1 if the rating is 4 or 5, otherwise 0.

Input Feature:

- Price

Train-Test Split:

80% training data and 20% testing data.

Model Result:

```
accuracy = accuracy_score(y_test, y_pred)

print("Model Accuracy:", round(accuracy * 100, 2), "%")

Model Accuracy: 58.5 %
```

The model achieved an accuracy of 58.5% on the test data.

Conclusion:

The result shows that price alone is not enough to accurately identify highly rated books. Other factors may be needed to improve the model's performance.

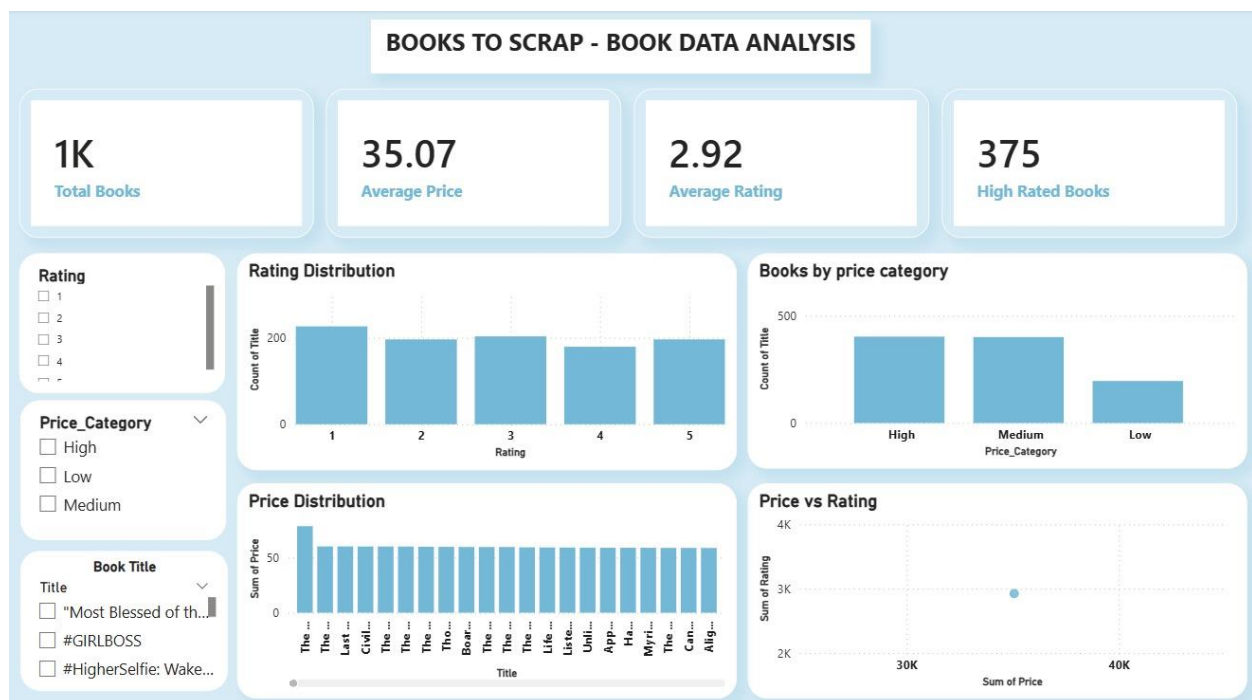
8. Power BI Dashboard

An interactive Power BI dashboard was created to present the book data in a simple and easy-to-understand way.

The main dashboard includes the following KPIs:

- Total Books
- Average Price
- Average Rating
- High-Rated Books

It also includes charts for rating distribution, price distribution, price categories, and the relationship between price and rating.

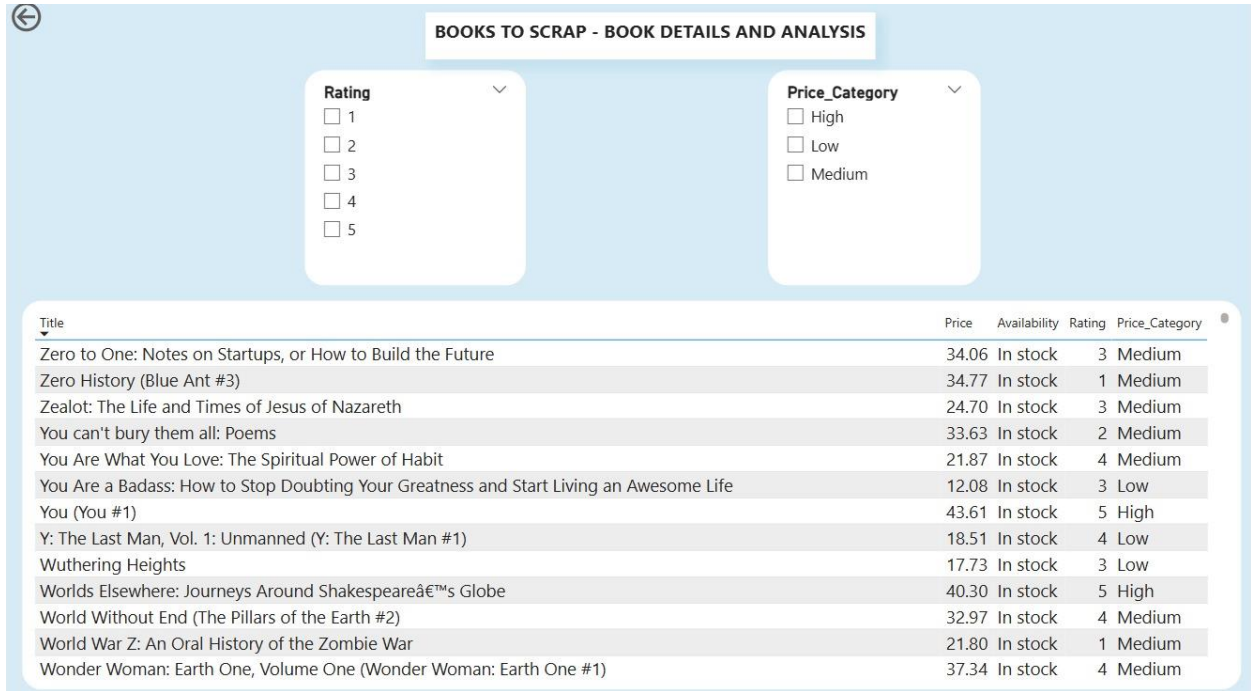


Key Finding:

The dashboard provides a quick overview of the books, their prices and ratings.

9. Book Details Dashboard

The second page provides detailed information about individual books. It can be accessed using the drill-through feature from the main dashboard.



Rating	Price_Category
☐ 1	☐ High
☐ 2	☐ Low
☐ 3	☐ Medium
☐ 4	
☐ 5	

Title	Price	Availability	Rating	Price_Category
Zero to One: Notes on Startups, or How to Build the Future	34.06	In stock	3	Medium
Zero History (Blue Ant #3)	34.77	In stock	1	Medium
Zealot: The Life and Times of Jesus of Nazareth	24.70	In stock	3	Medium
You can't bury them all: Poems	33.63	In stock	2	Medium
You Are What You Love: The Spiritual Power of Habit	21.87	In stock	4	Medium
You Are a Badass: How to Stop Doubting Your Greatness and Start Living an Awesome Life	12.08	In stock	3	Low
You (You #1)	43.61	In stock	5	High
Y: The Last Man, Vol. 1: Unmanned (Y: The Last Man #1)	18.51	In stock	4	Low
Wuthering Heights	17.73	In stock	3	Low
Worlds Elsewhere: Journeys Around Shakespeare's Globe	40.30	In stock	5	High
World Without End (The Pillars of the Earth #2)	32.97	In stock	4	Medium
World War Z: An Oral History of the Zombie War	21.80	In stock	1	Medium
Wonder Woman: Earth One, Volume One (Wonder Woman: Earth One #1)	37.34	In stock	4	Medium

Dashboard Features:

- Book details table
- Rating slicer
- Price Category slicer
- Drill-through
- Bookmark
- Sync slicer
- Row-Level Security (RLS)

Key Finding:

The second page makes it easier to view and filter individual book details based on the selected criteria.

10. Recommendations & Conclusion

Recommendations

- Price alone is not enough to predict a book's rating.
- Adding more relevant features could improve the model.
- The Power BI dashboard can be used to easily compare book prices and ratings.

Conclusion

In this project, I collected 1,000 book records from Books to Scrape using Python. I cleaned and analysed the data, created visualisations, and built a Decision Tree model.

The model achieved 58.5% accuracy. I also created a two-page Power BI dashboard to present the main results in an interactive way.

Overall, this project gave me practical experience in web scraping, data cleaning, EDA, machine learning and Power BI.

11. Project Video

The project video explains the complete project, including the problem statement, web scraping process, data cleaning, EDA, machine learning model, and Power BI dashboard.

Video Link:

<https://drive.google.com/file/d/1cC1aGUAu-6-UyjUAfUbfiAMvZ3lyXF0r/view?usp=sharing>